

Survival analysis

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- Goal

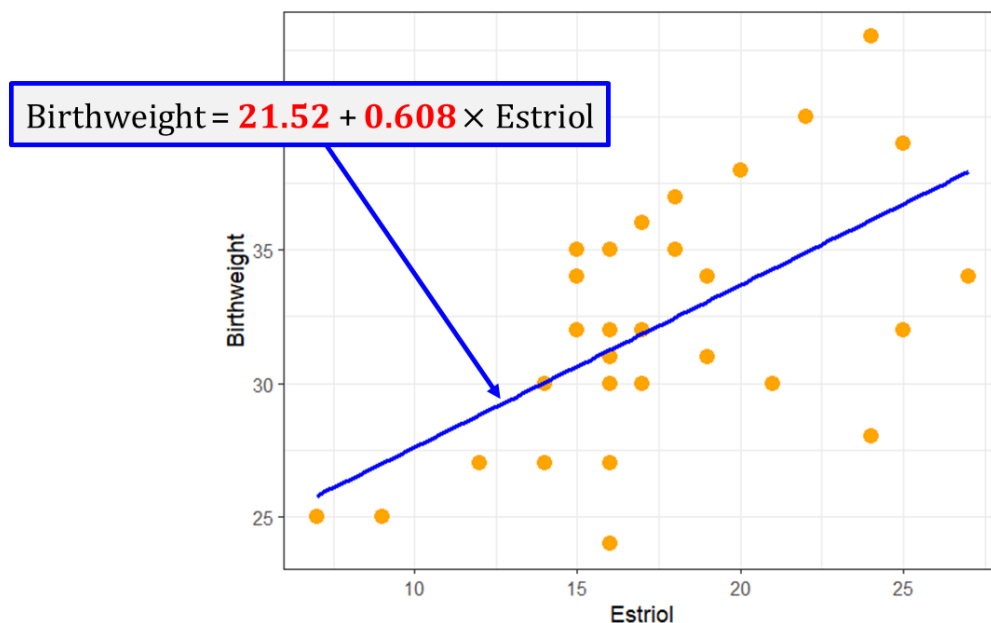
- 결과변수 (Y)와 설명변수 (X s)간의 **인과관계 (causal relationship)** 추론
- 인과관계를 $Y = f(X)$ 로 모형화

$$Y = f(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K) + \epsilon$$

- 결과변수를 설명변수들의 선형 조합 (linear combination) = 가중합 (weighted sum)의 함수로 모형화

- Objectives

- 모형의 추정, 유의성 확인, 설명력 평가
- 결과변수 설명/예측에 대한 각 설명변수의 상대적 기여도 추정
→ 위험인자 분석 (risk factor analysis)
: outcome에 영향을 미치는 설명변수는 무엇인가?
- 특정 설명변수 값에 대한 결과변수 값 예측
→ 예측 모형 (prediction modeling)
: 설명변수들로 outcome을 예측할 수 있는가?



- 일반화 선형모형 (Generalized linear model, GLM)
 - $g(Y) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \epsilon$ and **pre-chosen link function for $g(\cdot)$ according to type of outcome**
 - Assume non-normal error
- 로지스틱 회귀모형 (Logistic regression model)
 - 범주형 결과변수 (ex. **Control** vs **Case**, **Normal** vs **Disease**, **Class 1** vs **Class 2**)
 - GLM for binary outcome with binomial error

- **Logit link function** $g(Y) = \ln\left(\frac{p}{1-p}\right)$ where $p = P(\mathbf{Y} = \mathbf{1})$

- Assumption
 - Linear relationship between **log odds** $\ln\left(\frac{p}{1-p}\right)$ and $\mathbf{X}$

[Review] Odds Ratio vs Beta coefficient (β)

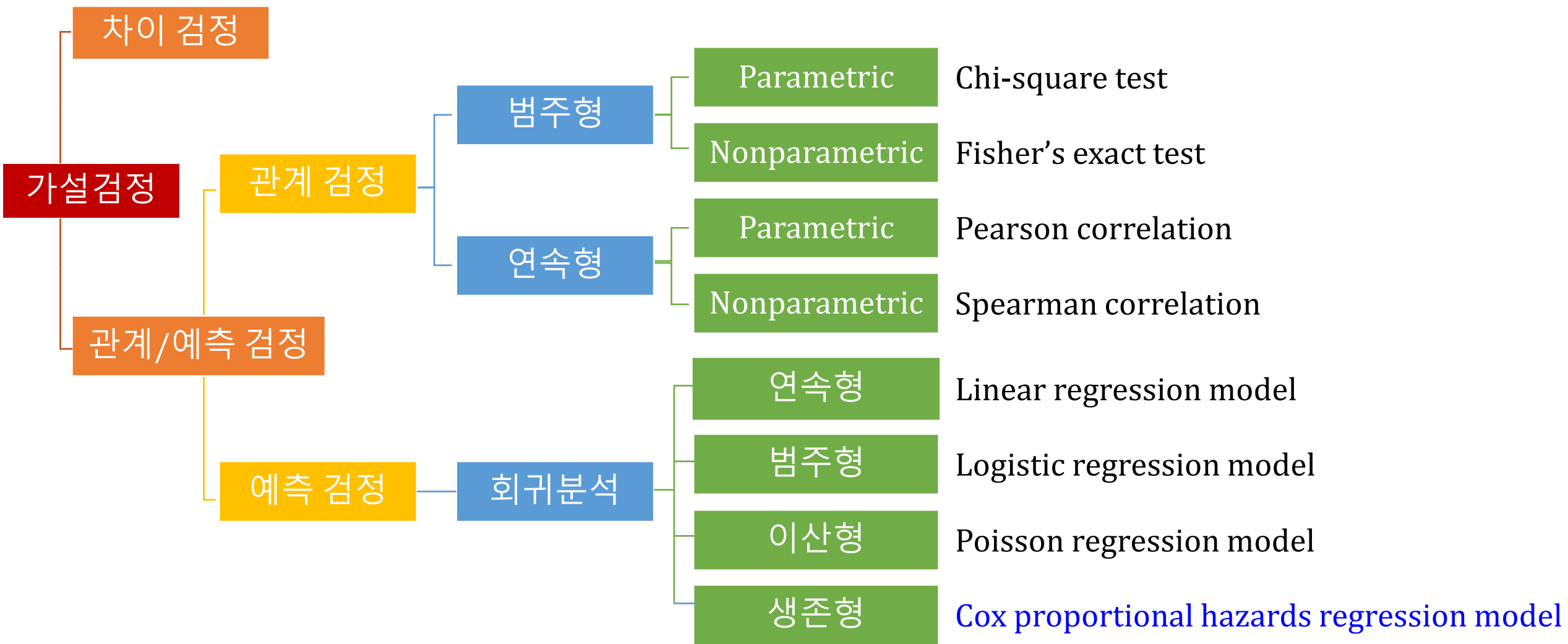
- 설명변수 별 오즈비 (OR) vs 회귀계수 (β)

$$OR_k = OR(X_k) = \frac{\Delta(X_k = 1)}{\Delta(X_k = 0)} = \exp(\beta_k)$$

$\Delta(\mathbf{p}) = \text{odds}$
 $\mathbf{p} = \mathbf{P}(Y = 1)$

$$\beta_k = \ln\{OR_k\} = \ln(\Delta(X_k = 1)) - \ln(\Delta(X_k = 0))$$

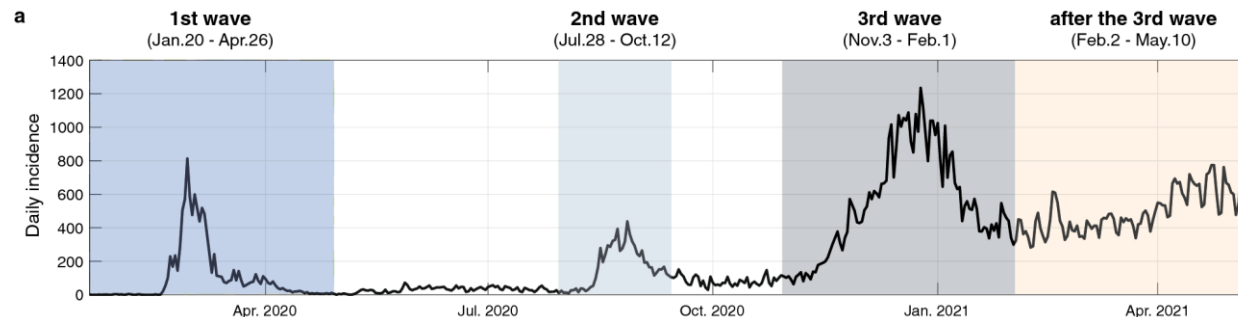
- X_k 값이 한 단위 증가할 때, case일 log odds가 β_k 만큼 증가
- X_k 값이 한 단위 증가할 때, case일 odds가 $\exp(\beta_k)$ 만큼 증가



- Cox regression model
 - Introduction to Survival analysis
 - Descriptive Statistics for Time-to-event data
 - Difference test for Survival analysis
 - Cox's proportional hazard model
 - R 실습

Introduction to Survival analysis (Time-to-Event analysis)

- 데이터 관측기간의 길이에 따라 구분
- Cross-sectional data (횡단 자료)
 - : 특정 시점 또는 짧은 기간동안 한번 수집한 자료
 - 예. 국민건강영양조사 자료
 - 특정 시점에서의 상태나 연관성 분석
 - 예. 2025년 3월 현재 고혈압 여부에 따른 비만율 분석
- Longitudinal data (종단 자료)
 - : 동일한 대상을 여러 기간에 걸쳐 측정하거나 여러 시점에 걸쳐 반복측정한 자료
 - 예. 건강보험공단 코호트 자료
 - 시간에 따른 변화나 인과관계 분석
 - 예. 2020년부터 2025년까지의 고혈압 여부에 따른 비만 위험도 분석
 - 예. 한국의 COVID-19 pandemic 동안의 일별 발생건수 패턴 변화 분석
 - Repeated measures data (반복측정 자료), Time-series data (시계열 자료), Survival data (생존 자료)



Kim, Sungchan, et al. "Discovering spatiotemporal patterns of COVID-19 pandemic in South Korea." *Scientific Reports* 11.1 (2021): 24470.



Original Investigation | Oncology

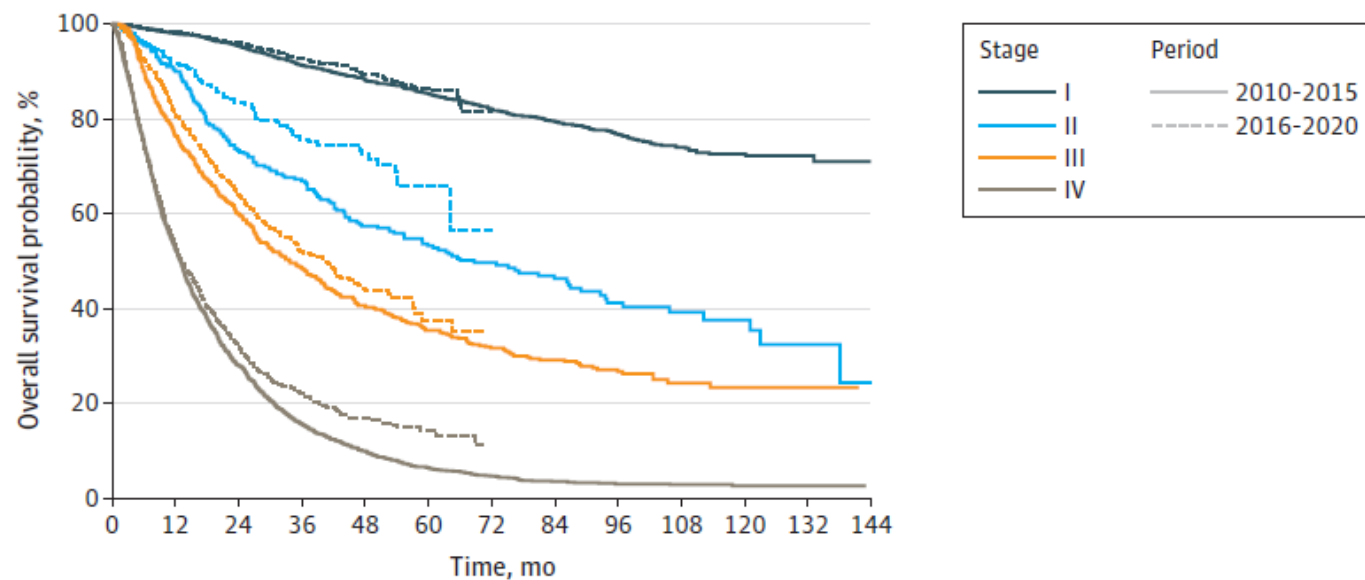
Trends in Survival Rates of Non-Small Cell Lung Cancer With Use of Molecular Testing and Targeted Therapy in Korea, 2010-2020

IMPORTANCE Over the past 10 years, treatment of non-small cell lung cancer (NSCLC) has been continually revolutionized. However, standard clinical trials may not reflect current multiple lines of treatment and corresponding outcomes in a timely manner.

OBJECTIVE To investigate outcomes associated with new treatment of NSCLC in a clinical setting.

DESIGN, SETTING, AND PARTICIPANTS This cohort study included patients with NSCLC between January 1, 2010, and November 30, 2020, who received any anticancer treatment at Samsung Medical Center in Korea. Data were analyzed from November 2021 through February 2022.

Figure 3. Overall Survival Curves for Patients With Adenocarcinoma (AD) and Without Druggable Variations



No. at risk

Stage I

2010-2015 1801 1724 1632 1522 1439 1301 1062 805 586 363 177 74 1

2016-2020 1819 1710 1257 861 541 223 0 0 0 0 0 0 0

Stage II

2010-2015 295 258 205 185 155 131 104 76 47 34 17 6 1

2016-2020 315 264 184 125 71 20 0 0 0 0 0 0 0

Stage III

2010-2015 523 394 307 244 203 173 140 97 64 30 12 4 0

2016-2020 550 418 258 156 73 28 0 0 0 0 0 0 0

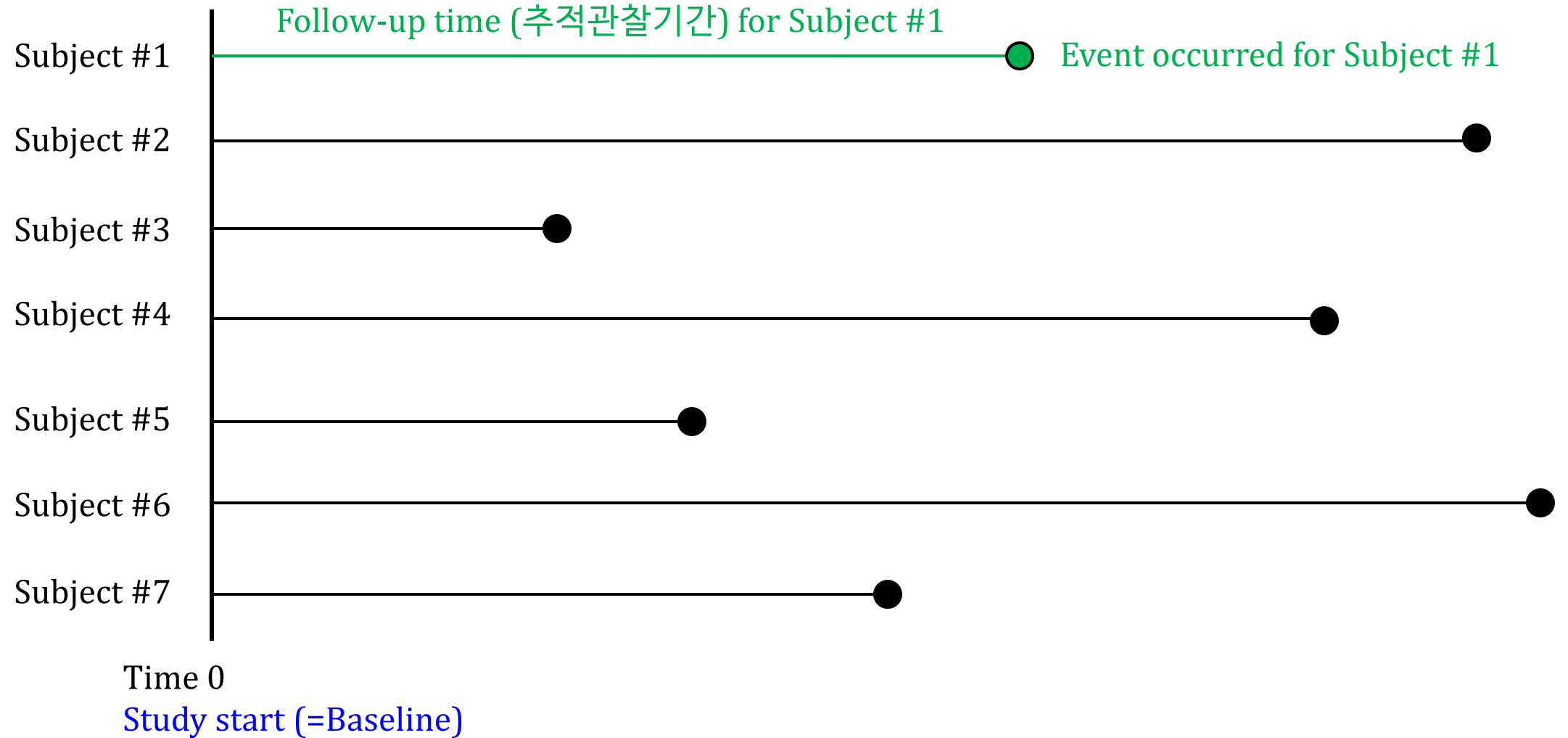
Stage IV

2010-2015 1432 748 395 218 138 86 59 37 21 16 9 4 0

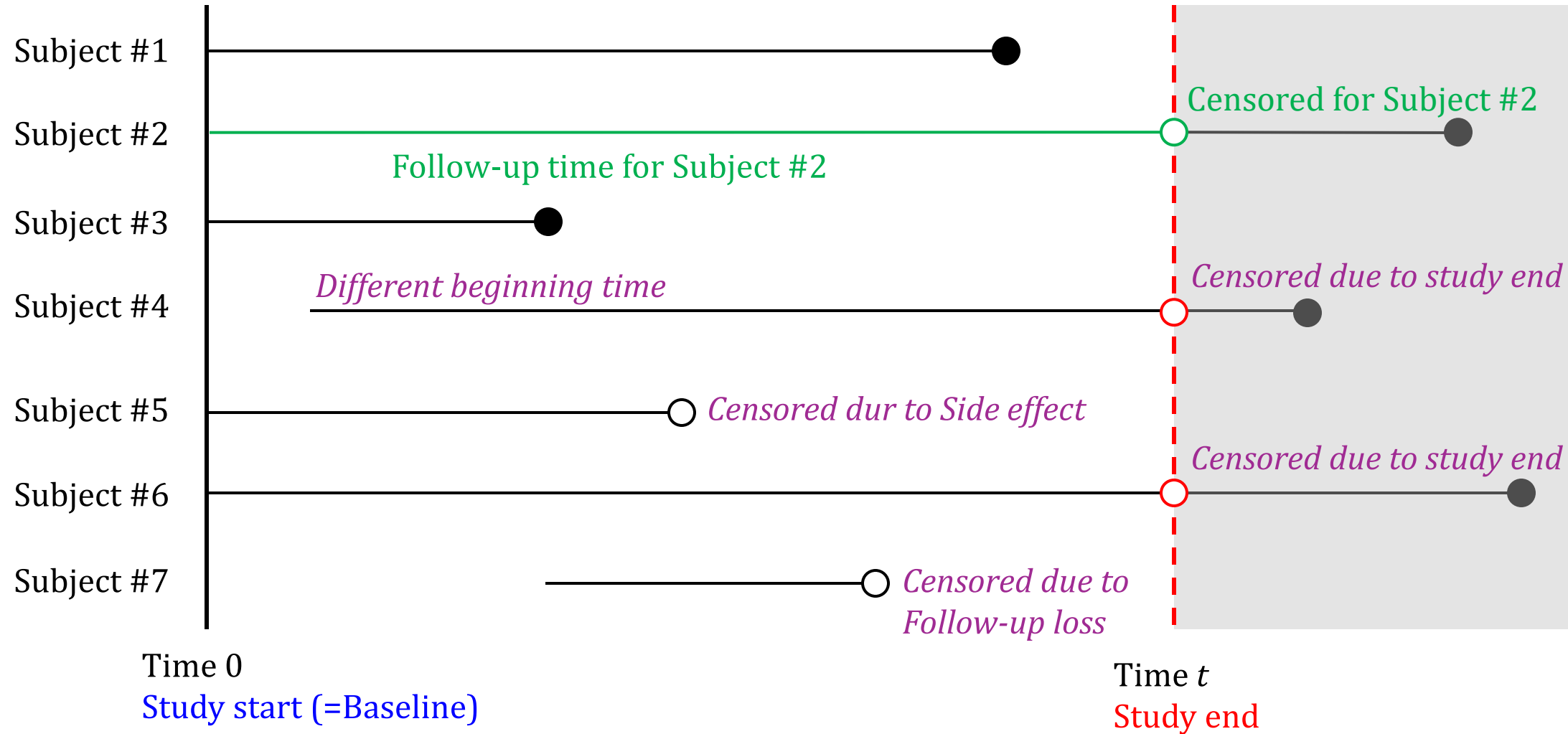
2016-2020 1120 545 258 128 57 18 0 0 0 0 0 0 0

시간에 따른 생존율의 감소 속도 (위험도) 확인

Follow-up Study (Ideal Setting)

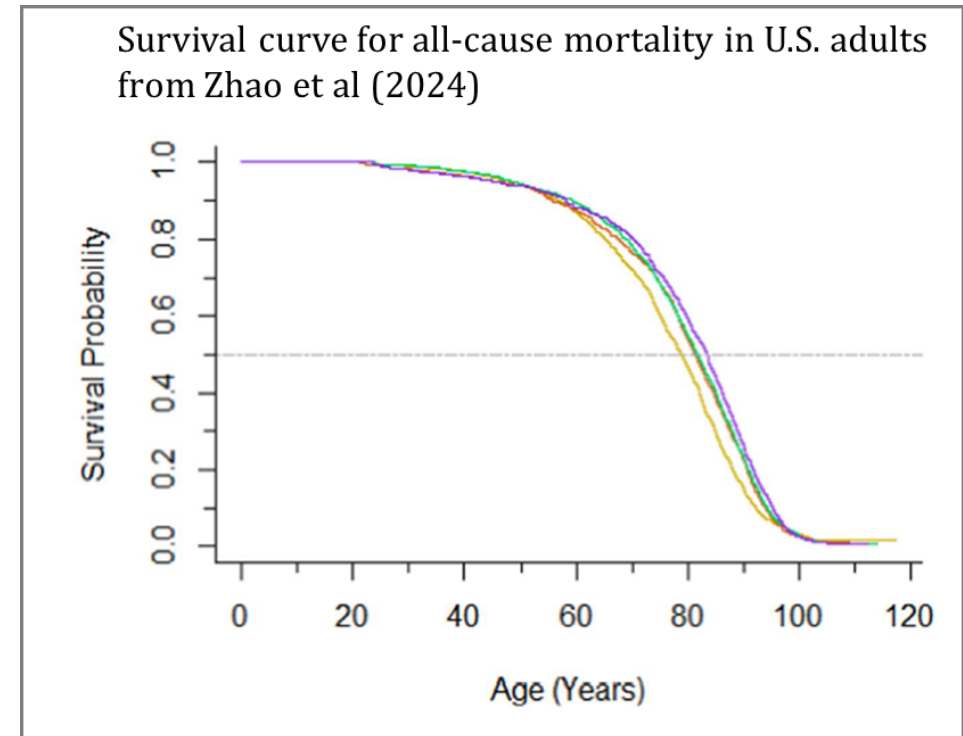


Follow-up Study (Realistic Setting)



- 시간에 따른 사망이나 재발 등의 변화를 관찰하는 분석
 - Event 발생 여부 (Event occurred or Censored)
 - Event 발생 또는 Censor까지 걸린 시간 (Follow-up time, Time-to-event)
- Censored data (중도절단 자료)
 - 다양한 이유로 사건이 발생하지 않아 사건발생까지 걸린 시간을 추정할 수 없는 자료
 - Possible reasons
 - Study end (termination of study)
 - Drop-out
 - Lost to follow-up
 - Withdrawn from study
 - Death from unrelated cause

- T = time-to-event
- Survival function
 - $S(t) = P(T > t)$: 시점 t 에서 살아 있을 확률 (Survival Probability)
- Hazard function
 - $h(t) = \frac{dS(t)}{dt} = \lim_{dt \rightarrow 0} \frac{P(t \leq T < t + dt \mid T \geq t)}{dt}$
: 시점 t 까지 살아 있을 때, 직후에 바로 event가 발생할 조건부 확률



Descriptive Statistics for Time-to-event data

- Survival probability at specific time t : 시점 t 에서 살아있을 확률 (ex. 5-year survival probability)

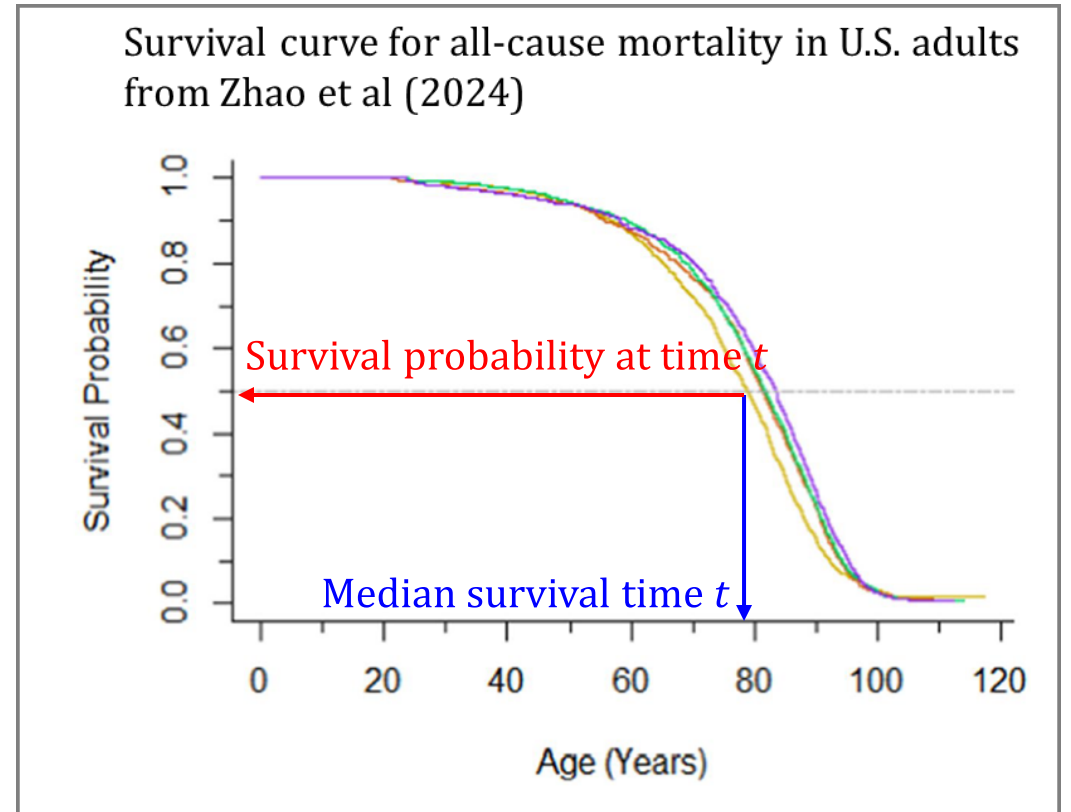
$$S(t) = P(T > t)$$

- Median survival time: 전체 생존 곡선의 중간점 (전체 환자의 50%가 생존했을 때의 시간)

$$t \quad \text{where } S(t) = 0.5$$

- Median follow-up time: 관찰된 전체 추적기간의 중앙값
 - Reverse Kaplan-Meier method
- Incidence rate: 단위 시간 당 사건 발생률 (person-year 기반)

$$\lambda(t) = \frac{\text{Number of events}}{\text{Total person-time}}$$



Difference test for Survival analysis

Framingham Heart Study라는 코호트 연구에서 과거 당뇨병 진단 여부 (DIABETES)와 심뇌혈관질환 (CVD) 발생 위험 간의 연관성을 확인하고자 한다.

이때 Period 1에 해당하는 대상군 (PERIOD=1)에 한하여,
전체 연구기간에서 당뇨병 발생여부에 따른 심뇌혈관질환 발생의 생존곡선이 다른지 여부를 통계적으로 검정하라.

가설

$H_0: S_1(t) = S_2(t)$ for all t (모든 시점에서 두 군의 생존확률은 같다)
 $H_1: S_1(t) \neq S_2(t)$ for some t

유의수준

0.05

검정법

양측 이표본 로그순위 검정 (two-sided two-sample log-rank test)

양측 (two-sided): 차이 검정

이표본 (two-sample): 2개의 독립 집단 간의 비교

Log-rank 검정 : 생존자료에 대한 검정통계량의 확률분포

```
# 데이터 가져오기
# install.packages("riskCommunicator")
library(riskCommunicator)
data(framingham)

# Period=1에 해당하는 군 선별 (Subgroup)
library(dplyr)
tmp <- framingham %>% dplyr::filter(PERIOD == 1)

# CVD 발생 비율 및 time-to-event 확인
with(tmp, table(DIABETES, CVD))
with(tmp, tapply(TIMECVD, DIABETES, summary))

# time-to-event=0인 환자군 삭제 (time-to-event는 항상 양의 값)
tmp <- tmp %>% dplyr::filter(TIMECVD > 0)

# CVD 발생 비율 및 time-to-event 확인
with(tmp, table(DIABETES, CVD))
with(tmp, tapply(TIMECVD, DIABETES, summary))
```

CVD		
DIABETES	0	1
0	3230	1083
1	47	74

\$`0`					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	5317	8766	6886	8766	8766
\$`1`					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	1710	4283	4392	6677	8766

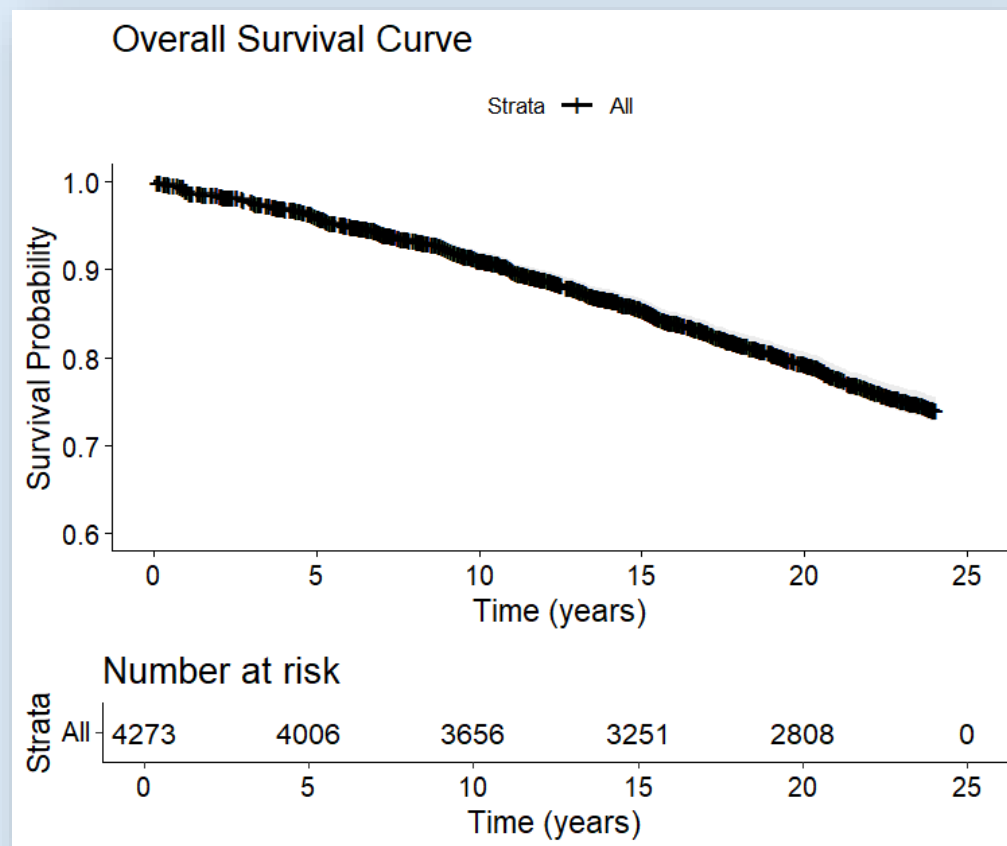
CVD		
DIABETES	0	1
0	3230	933
1	47	63

\$`0`						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
26	5726	8766	7134	8766	8766	
\$`1`						
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	
47	2654	4596	4831	7451	8766	

```
# Package for survival analysis
library(survival)

# ggplot Package for survival analysis
# install.packages("survminer")
library(survminer)

# Survival curve for total population
surv_fit <- survfit(Surv(TIMECVD/365, CVD) ~ 1, data=tmp)
ggsurvplot(surv_fit,
  data=tmp,
  conf.int = TRUE,
  risk.table = TRUE,
  palette = "black",
  ylim = c(0.6, 1),
  xlab = "Time (years)",
  ylab = "Survival Probability",
  title = "Overall Survival Curve")
```



Call: survfit(formula = Surv(TIMECVD/365, CVD) ~ 1, data = tmp)

time	n.risk	n.event	survival	std.err	lower	95% CI	upper	95% CI
5	4006	167	0.960	0.00300	0.955		0.966	
10	3656	202	0.911	0.00442	0.902		0.920	
15	3251	218	0.855	0.00554	0.845		0.866	
20	2808	228	0.793	0.00648	0.781		0.806	

해석

- CVD에 대한 5년 생존율 96%
- 5년째 살아있는 (CVD 발생하지 않은) 사람 4006명
- 5년내에 CVD 발생한 사람 167명

```
# Survival curve according to DIABETES
```

```
surv_fit <- survfit(Surv(TIMECVD/365, CVD) ~ DIABETES, data=tmp)
```

```
surv_fit
```

```
ggsurvplot(surv_fit,
```

```
  data=tmp,
```

```
  conf.int = TRUE,
```

```
  risk.table = TRUE,
```

```
  pval=TRUE,
```

```
  xlab = "Time (years)",
```

```
  ylab = "Survival Probability",
```

```
  title = "Overall Survival Curve according to DM")
```

```
# Log-rank test
```

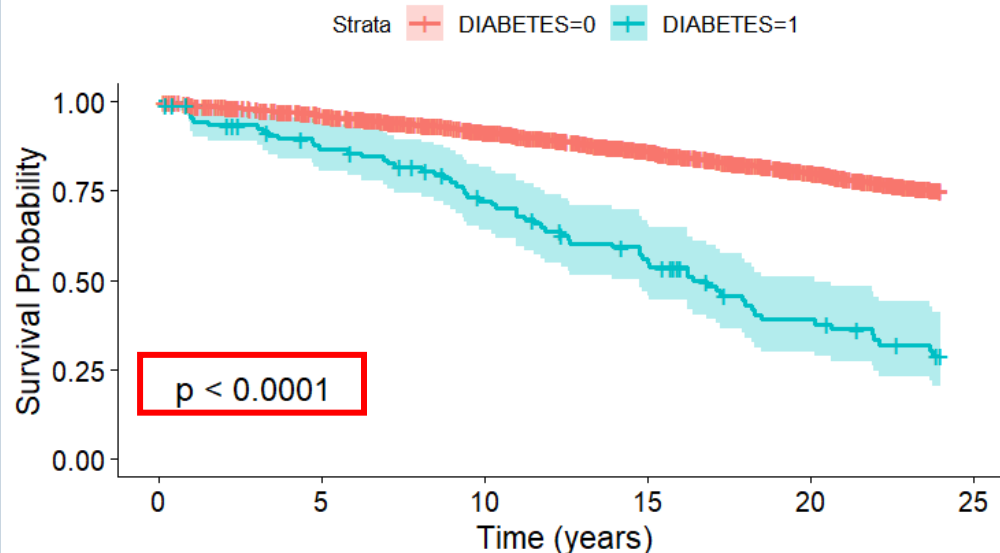
```
survdif(Surv(TIMECVD/365, CVD) ~ DIABETES, data=tmp)
```

결론

- 과거 당뇨병 진단여부에 따른 CVD 발생의 생존 곡선은 유의하게 다르다고 할 수 있다.

	n	events	median	0.95LCL	0.95UCL
DIABETES=0	4163	933	NA	NA	NA
DIABETES=1	110	63	16.4	14.7	20.1

Overall Survival Curve according to DM



Number at risk

Strata	DIABETES=0	DIABETES=1	0	5	10	15	20	25
DIABETES=0	4163	3918	3588	3202	2779	0		
DIABETES=1	110	88	68	49	29	0		

Time (years)

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
DIABETES=0	4163	933	979.8	2.24	138
DIABETES=1	110	63	16.2	135.58	138

Chisq= 138 on 1 degrees of freedom, $p = <2e-16$

Cox's Proportional Hazard Model

- 일반화 선형모형 (Generalized linear model, GLM)
 - $g(Y) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \epsilon$ and **pre-chosen link function for $g(\cdot)$ according to type of outcome**
 - Assume non-normal error
- 콕스 비례위험 회귀모형 (Cox's proportional hazard regression model)
 - **생존형 결과변수** (time-to-event data)
 - GLM for time-to-event outcome

- **Log hazard link function**

$$\log(h(t)) = \log\left(\frac{dS(t)}{dt}\right) \quad \text{where } S(t) = P(T > t)$$

- Assumption
 - Linear relationship between **log hazard** $\log(h(t))$ and X
 - Proportional hazard assumption (PH assumption, 비례 위험 가정)
 - 위험도의 비(Hazard ratio)는 모든 시간에서 일정하다
 - 어느 시점이던 두 집단 간 사건이 발생할 위험의 비율이 일정하게 유지된다
 - Shoenfeld residuals test (Shoenfeld individual test)

$$\frac{h(t | X_k = 1)}{h(t | X_k = 0)} = c \text{ (constant)}$$

	Linear regression	Logistic regression	Cox's PH regression
모형의 유의성	F-test	Chi-square test, AIC	LRT, Wald test, Score test
모형의 설명력	R^2	Cox & Snell R^2 Nagelkerke R^2	Cox & Snell R^2 Nagelkerke R^2 Concordance index (C-index)
회귀계수 (β)의 유의성	t-test	LRT, Score test, Wald test	LRT, Score test, Wald test
회귀계수 (β)의 해석 (by increment of X)	Increment in outcome Y	Increment in log odds	Increment in log hazard

AIC: Akai information criterion
LRT: Log likelihood test

- Survival Probability

$$S(t) = P(T > t)$$

- Hazard function

$$h(t) = \frac{dS(t)}{dt} = \lim_{dt \rightarrow 0} \frac{P(t \leq T < t + dt \mid T \geq t)}{dt}$$

- Log hazard

$$\ln(h(t)) = \log\left(\frac{dS(t)}{dt}\right) = \beta_0 + \sum_{k=1}^K \beta_k X_k$$

- Hazard ratio

$$HR(X_k) = \frac{h(t \mid X_k = 1)}{h(t \mid X_k = 0)} = \exp(\beta_k)$$

- Beta coefficient

$$\beta_k = \ln\{HR(X_k)\}$$

- 설명변수 별 위험비 (HR) vs 회귀계수 (β)

$$HR_k = HR(X_k) = \frac{h(t | X_k = 1)}{h(t | X_k = 0)} = \exp(\beta_k)$$

$$\beta_k = \ln\{HR_k\} = \ln(h(t | X_k = 1)) - \ln(h(t | X_k = 0))$$

- X_k 값이 한 단위 증가할 때, case가 발생할 log hazard가 β_k 만큼 증가
- X_k 값이 한 단위 증가할 때, case가 발생할 hazard가 $\exp(\beta_k)$ 만큼 증가

- 회귀계수 β_k 의 의미

$$H_0 : \beta_k = 0 \leftrightarrow \text{HR}_k = 1 \leftrightarrow h(t | X_k = 1) = h(t | X_k = 0)$$

- No difference in hazard
- No association between Y & X_k

$$H_0 : \beta_k \neq 0 \leftrightarrow \text{HR}_k \neq 1 \leftrightarrow h(t | X_k = 1) \neq h(t | X_k = 0)$$

- Significant difference in hazard
- Significant association between Y & X_k

- 회귀 계수 β_k 의 부호

- $\beta_k > 0 \leftrightarrow \text{HR}(X_k) > 1 \leftrightarrow h(t | X_k = 1) > h(t | X_k = 0)$: $X_k = 1$ 일때 odds $\uparrow$ (risky)
- $\beta_k < 0 \leftrightarrow \text{HR}(X_k) < 1 \leftrightarrow h(t | X_k = 1) < h(t | X_k = 0)$: $X_k = 1$ 일때 odds $\downarrow$ (protective)

- 회귀 계수 β_k 의 절대값

- 회귀 계수의 절대값이 클수록 해당 설명변수의 영향력/예측력이 크다는 의미
- HR값이 1과 차이가 많이 날수록** 해당 설명변수의 영향력/예측력이 크다는 의미

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Added Value of Pelvic CT after Treatment of HCC

Arim Yeom, MD • Sang Ah Chi, BS • Kyoung Doo Song, MD

Background: In patients with hepatocellular carcinoma (HCC) who undergo follow-up with CT after treatment, the benefit of routinely including pelvic coverage is not well substantiated.

Purpose: To investigate the added value of pelvic coverage at follow-up liver CT in detecting pelvic metastasis or incidental findings in patients treated for HCC.

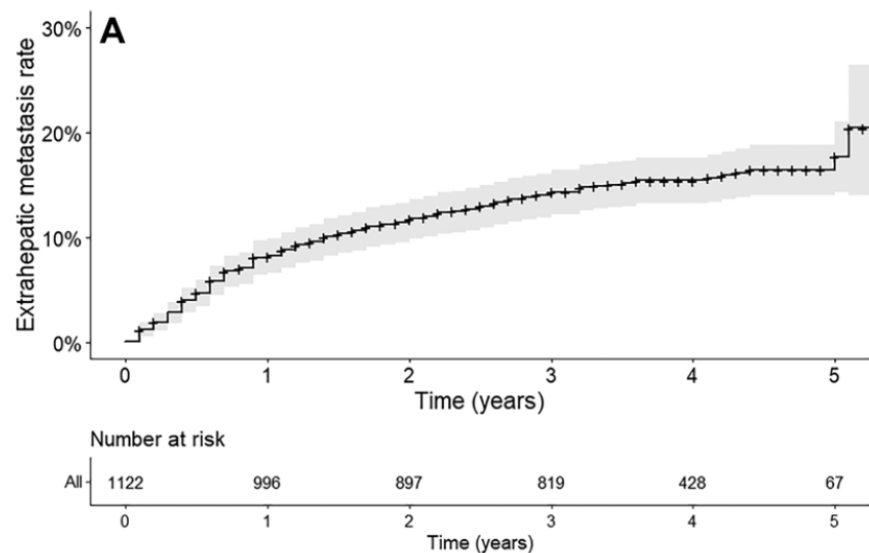
Materials and Methods: This retrospective study included patients who were diagnosed with HCC between January 2016 and December 2017 and followed up with liver CT after treatment. Cumulative rates of extrahepatic metastasis, isolated pelvic metastasis, and incidental pelvic tumor were estimated by using the Kaplan-Meier method. Cox proportional hazard models were used to identify risk factors for extrahepatic and isolated pelvic metastases. Radiation dose from pelvic coverage was also calculated.

Statistical Analysis

Cumulative rates of extrahepatic metastasis, isolated pelvic metastasis, and incidental pelvic tumors were estimated using the Kaplan-Meier method. Log rank tests were used to compare curves between groups of categorical variables. Median follow-up duration was calculated by the reverse

Kaplan-Meier estimation method. Cox proportional hazard models were used to identify factors associated with an increased risk of extrahepatic and isolated pelvic metastases. Multivariable Cox regression analysis was performed using variables selected from backward elimination based on the Akaike information criterion. For backward elimination in this study, if deleting a term lowered Akaike information criterion, the term that lowered Akaike information criterion the most when deleted was eliminated from the model. The final model was determined when there was no term to be eligible for removal (ie, no further decrement of Akaike information criterion was made when deleting one of the terms in the model). The proportional hazards assumption was checked using a test on Schoenfeld residuals and was found to be valid for all factors investigated. The

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Cumulative Incidence Rates of Lesions at Pelvic CT

The cumulative rates of extrahepatic metastasis at 1, 2, and 3 years were 8.3% (95% CI: 6.6, 9.9), 11.8% (95% CI: 9.9, 13.7), and 14.4% (95% CI: 12.2, 16.4), respectively. The cumulative rates of isolated pelvic metastasis at 1, 2, and 3 years were 0.8% (95% CI: 0.2, 1.3), 1.1% (95% CI: 0.4, 1.7), and 1.4% (95% CI: 0.7, 2.2), respectively. The cumulative rates of incidental pelvic tumors at 1, 2, and 3 years were 0.2% (95% CI: 0, 0.4), 0.4% (95% CI: 0, 0.8), and 0.5% (95% CI: 0.1, 1.0), respectively (Fig 2).

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Multivariable analysis showed that baseline patient factors were associated with an increased risk of extrahepatic metastasis, as follows: protein induced by vitamin K absence or antagonist-II (hazard ratio [HR], 1.2; 95% CI: 1.1, 1.3; $P = .001$), size of the largest tumor (HR for >3 cm and ≤5 cm vs ≤3 cm, 1.0 [95% CI: 0.6, 1.8; $P > .99$]; HR for >5 cm vs ≤3 cm, 1.8 [95% CI: 1.0, 3.2; $P = .03$ after Bonferroni correction]), T stage (HR for T2 vs T1, 2.0 [95% CI: 0.9, 4.5; $P = .13$]; HR for T3 vs T1, 2.7 [95% CI: 1.1, 6.6; $P = .03$]; HR for T4 vs T1, 4.2 [95% CI: 1.5, 11.9; $P = .003$ after Bonferroni correction]), and initial treatment method (HR, 2.4; 95% CI: 1.6, 3.5; $P < .001$) (Table 3).

Table 3: Univariable and Multivariable Cox Regression Analyses of Patient and Tumor Characteristics Associated with Extrahepatic Metastasis

Parameter	Univariable Analysis		Multivariable Analysis	
	HR	<i>P</i> Value	HR	<i>P</i> Value
Age	1.0 (1.0, 1.0)	.49	1.0 (1.0, 1.0)	.11
Sex (reference: female sex)		.15		
Male sex	1.4 (0.9, 2.1)			
Etiology (reference: HBV)*		.90		
HBV and HCV	0.6 (0.1, 6.8)	>.99		
HCV	1.2 (0.7, 2.3)	>.99		
Alcohol	1.0 (0.5, 2.0)	>.99		
Others	1.0 (0.6, 1.9)	>.99		
Platelet†	1.5 (1.1, 2.1)	.02		
Bilirubin†	1.1 (0.8, 1.4)	.60		
AST†	2.0 (1.6, 2.6)	<.001		
ALT†	1.4 (1.1, 1.8)	.003		
Albumin	0.9 (0.7, 1.2)	.58		
Prothrombin time	0.7 (0.2, 2.4)	.54		
AFP†	1.2 (1.2, 1.3)	<.001		
PIVKA-II†	1.4 (1.3, 1.5)	<.001	1.2 (1.1, 1.3)	.001
Child-Pugh class (reference: A)*		.71		
B	1.2 (0.6, 2.5)	>.99		
C	0.6 (0.1, 5.5)	>.99		
No. of tumors (reference: 1)*		<.001		
2	1.7 (1.1, 2.8)	.02		
3	2.4 (1.2, 4.7)	.008		
≥4	5.1 (3.2, 8.2)	<.001		
Tumor size (reference: ≤3 cm)*		<.001		.02
3–5 cm	1.8 (1.1, 3.1)	.03	1.0 (0.6, 1.8)	>.99
>5 cm	6.4 (4.3, 9.5)	<.001	1.8 (1.0, 3.2)	.03
T stage (reference: T1)*		<.001		.008
T2	3.3 (1.6, 7.1)	<.001	2.0 (0.9, 4.5)	.13
T3	9.7 (4.6, 20.6)	<.001	2.7 (1.1, 6.6)	.03
T4	24.4 (10.7, 55.8)	<.001	4.2 (1.5, 11.9)	.003
BCLC (reference: 0)*		<.001		
A	2.8 (1.3, 6.2)	.004		
B	9.1 (3.9, 20.9)	<.001		
C	10.9 (4.9, 24.0)	<.001		
D	2.2 (0.2, 29.9)	>.99		
Initial treatment (reference: curative)		<.001		<.001
Palliative	4.7 (3.4, 6.4)	<.001	2.4 (1.6, 3.5)	

- Demographic data of 857 patients with ACS

변수명	설명
age	환자 나이
sex	성별 ("Male" or "Female")
cardiogenicShock	심장성 쇼크 ("No" or "Yes")
entry	혈관 접근 경로 ("Femoral" or "Radial")
Dx	최종 진단명 ("STEMI", "NSTEMI" or "Unstable Angina")
EF	심장 초음파로 측정한 박출률 (%)
height	신장 (cm)
weight	체중 (kg)
BMI	체질량지수 (body mass index, kg/m2)
obesity	비만 여부 ("No" or "Yes")
TC	총콜레스테롤 (total cholesterol level, mg/dL)
LDLC	LDL 콜레스테롤 수치 (mg/dL)
HDLC	HDL 콜레스테롤 수치 (mg/dL)
TG	중성지방 농도 (triglyceride level, mg/dL)
DM	당뇨병 진단 여부 ("No" or "Yes")
HBP	고혈압 진단 여부 ("No" or "Yes")
smoking	흡연의 과거력 ("Never", "Ex-smoker" or "Smoker")

```
# 당뇨병과 CVD 발생위험간의 연관성 분석 - Univariable analysis
surv_fit <- coxph(Surv(TIMECVD/365, CVD) ~ DIABETES, data=tmp)
summary(surv_fit)
extractHR(surv_fit)

# PH 가정 확인
cox.zph(surv_fit)
ggcoxzph(cox.zph(surv_fit))
```

	HR	lcl	uc1	p
DIABETES	4.11	3.18	5.31	0

```
Call:
coxph(formula = Surv(TIMECVD/365, CVD) ~ DIABETES, data = tmp)

n= 4273, number of events= 996
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
DIABETES	1.4140	4.1125	0.1306	10.83	<2e-16 ***

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

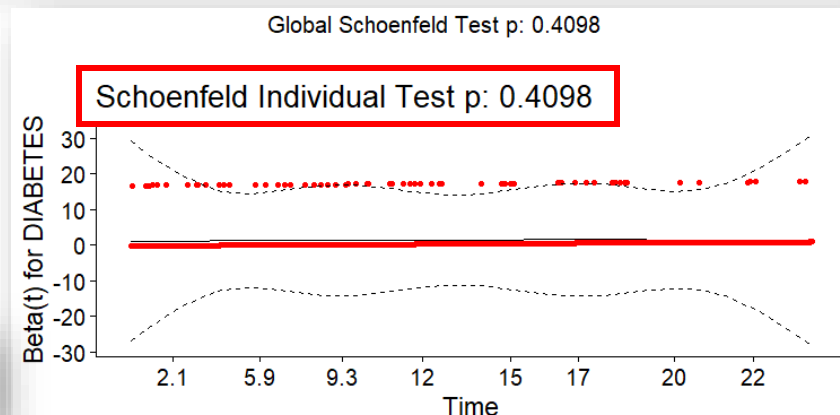
exp(coef) exp(-coef) lower .95 upper .95
DIABETES  4.112  0.2432    3.184    5.312

Concordance= 0.524 (se = 0.004 )
Likelihood ratio test= 80.2 on 1 df,  p=<2e-16
Wald test               = 117.2 on 1 df,  p=<2e-16
Score (logrank) test = 138.1 on 1 df,  p=<2e-16
```

결론

- $\widehat{HR}_{DM} = 4.11, p\text{-value} < 0.05$
 - 당뇨병을 진단받은 경우 CVD가 발생한 위험도(hazard)가 약 4.11배로 높아짐
 - 당뇨병 진단여부는 CVD 발생에 유의한 효과를 가짐
- 비례위험 가정을 만족함 ($P\text{-value} \geq 0.05$)

	chisq	df	p
DIABETES	0.679	1	0.41
GLOBAL	0.679	1	0.41



당뇨병과 CVD 발생위험간의 연관성 분석 - Multivariable analysis

```
surv_fit <- coxph(Surv(TIMECVD/365, CVD) ~ DIABETES + factor(SEX) + AGE, data=tmp)
```

```
summary(surv_fit)
```

```
extractHR(surv_fit)
```

PH 가정 확인

```
cox.zph(surv_fit)
```

```
ggcoxzph(cox.zph(surv_fit))
```

	coef	exp(coef)	se(coef)	z	Pr(> z)	
DIABETES	1.048637	2.853757	0.131809	7.956	1.78e-15	***
factor(SEX)2	-0.799220	0.449680	0.064480	-12.395	< 2e-16	***
AGE	0.064421	1.066542	0.003828	16.827	< 2e-16	***

	chisq	df	p
DIABETES	1.08427	1	0.30
factor(SEX)	0.06797	1	0.79
AGE	0.00172	1	0.97
GLOBAL	1.19539	3	0.75

	HR	lcl	ucl	p
DIABETES	2.85	2.20	3.69	0
factor(SEX)2	0.45	0.40	0.51	0
AGE	1.07	1.06	1.07	0

결론

- $\widehat{HR}_{diabetes} = 2.85$, $p\text{-value} < 0.05$
 - 나이와 성별이 일정하다고 가정할 때, 당뇨병을 진단받은 경우 CVD가 발생한 위험도가 약 2.85배로 높아짐
- $\widehat{HR}_{sex} = 0.45$, $p\text{-value} < 0.05$
 - 나이와 당뇨병 진단 여부가 일정하다고 가정할 때, 여자인 경우 CVD가 발생한 위험도가 약 0.45배로 낮아짐
- $\widehat{HR}_{age} = 1.07$, $p\text{-value} < 0.05$
 - 성별과 당뇨병 진단 여부가 일정하다고 가정할 때, 나이가 1단위 많은 경우 CVD가 발생한 위험도가 약 1.07배로 높아짐

R 실습

```
1. “framingham” 자료 가져오기  
install.packages("riskCommunicator")  
library(riskCommunicator)  
data(framingham)
```

2. Period 1에 해당하는 대상군 (PERIOD=1)에 한하여,
당뇨여부(DIABETES)로 심뇌혈관질환(CVD) 발생 위험을 예측할 수 있는 회귀식을 추정하고 추정된 위험비를 해석하라.

3. Period 1에 해당하는 대상군 (PERIOD=1)에 한하여,
성별(SEX), 나이(AGE), BMI, 수축기혈압(SYSBP), 심박수(HEARTRTE), 총콜레스테롤(TOTCHOL), 흡연 여부(CURSMOKE),
과거 고혈압 진단 여부(PREVBHYP) 및 항고혈압 약 복용 여부 (BPMEDS)를 보정한 후,
당뇨여부(DIABETES)로 심뇌혈관질환(CVD) 발생 위험을 예측할 수 있는 회귀식을 추정하고 추정된 위험비를 해석하라.

4. 3번의 추정된 회귀식의 다중공선성과 비례위험 가정을 확인하고 해석하라.

If you have ANY question,
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